

Motor Datasheet:

maxon2420C
20260721 081916 base7

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	93.0	mm
Machine Inner Diameter	40.8	mm
Machine Thickness / Width	26.9	mm
Stator Height	26.9	mm
Magnet Height	2.8	mm
Tooth Height	13.8	mm
Air Gap	0.50	mm
Fill Factor (input)	0.60	-
Slot Fill Factor (calculated)	Not calculated	-
Slot Width (tooth to tooth)	3.486	mm
Available Coil Space	1.743	mm
Assumed Coil Width (simulation)	1.543	mm
Assumed Coil Height (simulation)	13.613	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	951.9	g
Rotor Mass	370.0	g
Stator Mass	581.9	g
Total Copper Mass (pure copper)	165.8	g
Total Stator Steel Mass	416.1	g

Total Magnet Mass	95.1	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	4691.5	g·cm ²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	37.805	A
Supply Current (Peak)	53.465	A
Id (Peak)	-0.000	A
Iq (Peak)	53.465	A
Phase Advance Angle	0.0	°elec
Phase Resistance at operational temperature	16.456	mOhm
D-Axis Inductance (Ld)	41.674	μH
Q-Axis Inductance (Lq)	44.140	μH
Terminal Inductance (Ltt = Ld + Lq)	85.814	μH
PM Flux Linkage	6.976	mWb
Frequency at operating point (electrical)	500.000	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.000	V
Power Factor	0.910	-

Performance

Parameter	Value	Unit
Torque (Continuous)	5.61	Nm
Speed	3000.0	rpm
Dimensionless motor constant	0.0	-
BackEMF Constant (Ke, phase peak)	7.3	mV/rpm
Motor Constant (Km, only resistive losses)	668.3	mNm/√W
Specific Motor Constant (Km / mass)	702.0	mNm/√W/kg
Torque Constant (Kt, peak, no-load)	108.4	mNm/A

Torque Ripple	3.8	%
Saturation	3.1	%

Losses

Parameter	Value	Unit
Total Losses	20254687317559295227069610416387117022654909501144495961167667763609	
Resistive Losses	70.6	W
Iron Core Losses	20254687317559295227069610416387117022654909501144495961167667763609	
Magnet Eddy Current Losses	9.6	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

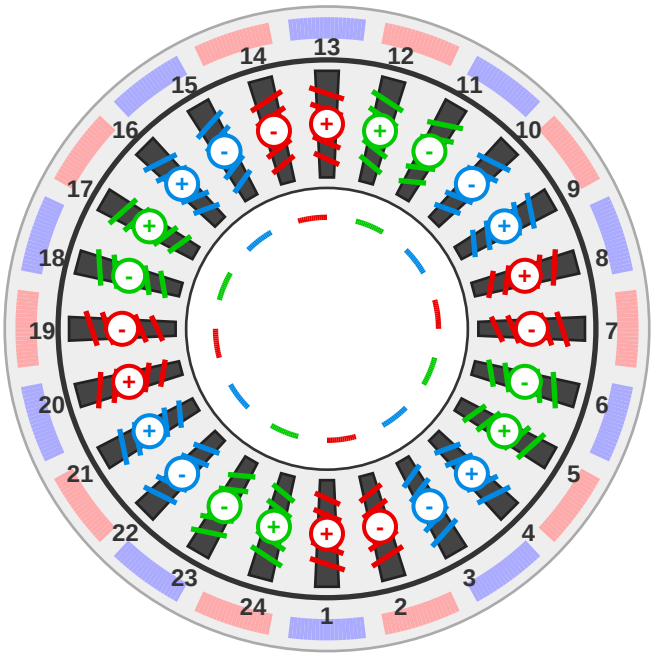
Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	24	-
Pole Pairs	10	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	5.0	-
Number of Phase Turns (all phases in series)	40.0	-

Winding Configuration

Schema: **AabBCcaABbcCAabBCcaABbcC**

Concentrated Winding | Balanced | Cogging: 120/rev | Ripple: 120/rev



Materials

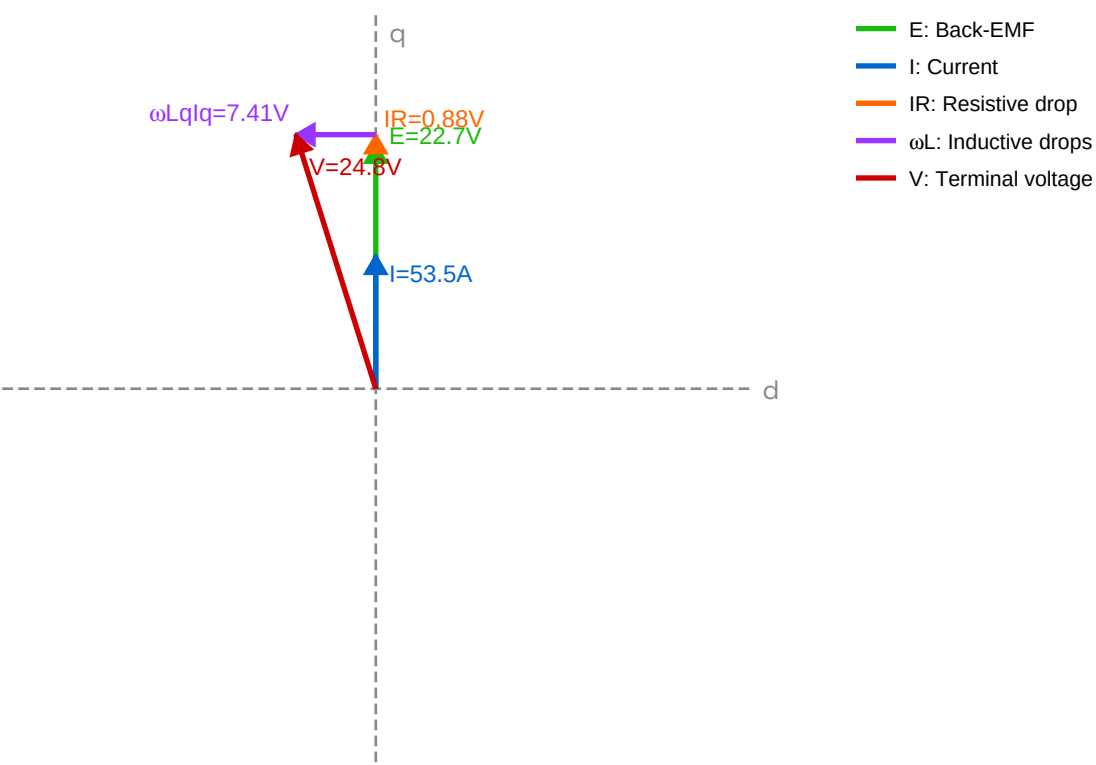
Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-
Rotor Backiron	11SMn30	-
Coil	Copper	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	22.700	V
Resistive Drop (Phase, Peak)	0.880	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	7.414	V

Inductive Drop Total (Peak)	7.414	V
Terminal Voltage (Phase, Peak)	24.761	V
Back-EMF Voltage (Line-to-Line, Peak)	39.317	V
Resistive Drop (Line-to-Line, Peak)	1.524	V
Inductive Drop (Line-to-Line, Peak)	12.841	V
Terminal Voltage (Line-to-Line, Peak)	42.887	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Mechanical Properties

Parameter	Value	Unit
Airgap Axial Force (Average)	881.44	N
Magnet Axial Force Positive Maximum	21.72	N
Magnet Axial Force Positive Minimum	0.03	N
Magnet Axial Force Negative Maximum	-16.32	N
Magnet Axial Force Negative Minimum	-0.01	N

Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.